Diploma in Cybersecurity
Cybersecurity Data Analysis Project

Unitec Institute of Technology

Course Coordinators

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Project Overview

In this assignment, you will undertake a data analysis project focused on cybersecurity. The project involves identifying a dataset, preferably related to cybersecurity operations and management, and conducting analyses to address specific cybersecurity challenges or questions. The project is divided into three parts:

- **Part 1: Project Intent Proposal** (20%)
- **Part 2: Final Report and Analysis** (60%)
- **Part 3: Recorded Presentation** (20%)

This project accounts for 60% of the total grade for the course.

Part 1: Project Intent Proposal (20%)

- **Dataset Identification:** Select a dataset, preferably related to cybersecurity, and provide a brief description.
- **Cybersecurity Questions or Problems:** Identify at least three relevant cybersecurity questions or problems.
- **Preliminary Analysis Plan:** Outline the types of analyses you plan to conduct.
- **Project Timeline:** Provide a timeline for project stages.

Submission: A written proposal (1000-1500 words) detailing the above requirements.

Part 2: Final Report and Analysis (60%)

- **Introduction:** Restate the cybersecurity questions and describe the dataset's relevance.
- **Descriptive and Exploratory Analysis:** Summarize the dataset and conduct exploratory data analysis.
- **Predictive Analysis (Optional):** Apply predictive analytics if relevant.
- **Conclusion:** Summarize key findings and their impact on cybersecurity practices.

Submission: A comprehensive report (3000-4000 words) with all visualizations, results, and a link to the Jupyter Notebook or Google Colab notebook containing all code.

Part 3: Recorded Presentation (20%)

- **Presentation Content:** Summarize the dataset, questions, process, findings, and recommendations.
- **Presentation Format:** 10-minute PowerPoint with voice-over narration.

Submission: A recorded presentation video file or link.

Evaluation Rubric

Criteria	**Description**	**Weight (%)**
	Part 1: Project Intent Proposal (20%)	
Clarity and Relevance of Dataset	Dataset choice is clear, relevant to cybersecurity, and well-described.	5
Significance of Cybersecurity Questions or Problems	Cybersecurity questions are significant and well-justified.	5
Feasibility and Thoroughness of Analysis Plan	The preliminary analysis plan is well-structured, feasible, and thorough.	5
Realistic and Detailed Project Timeline	The project timeline is realistic and detailed.	5
	Part 2: Final Report and Analysis (60%)	
Quality and Depth of Descriptive Analysis	Descriptive analysis is comprehensive, with clear statistics and visualizations.	15
Insightfulness and Rigor of Exploratory Analysis	Exploratory analysis effectively identifies patterns, correlations, and anomalies.	15
Appropriateness and Accuracy of Predictive Analysis (Optional)	Predictive analysis is well-executed with accurate models and clear explanations.	15
Quality of Explanation and Reflections	Explanations reflect a deep understanding of findings and their implications.	15
Clarity and Organization of Notebook	Code and visualizations in the notebook are well-organized and clear.	10
Clarity of Writing and Presentation	Writing is clear, organized, and supports the report's conclusions.	5
	Part 3: Recorded Presentation (20%)	
Clarity and Organization of Content	Presentation is organized, clearly summarizing the project and findings.	7
Effectiveness in Communicating Findings	Key findings and insights are effectively communicated.	7
Professionalism and Presentation Quality	Presentation is professional, with good audio-visual quality.	6

Table 1: Grading Rubric for Cybersecurity Data Analysis Project

Tools and Resources

- **Jupyter Notebook or Google Colab:** Use these platforms for data analysis, visualizations, and coding.

- **Data Sources:** Public datasets from sources like Kaggle, UCI Machine Learning Repository, or other reputable data repositories.

Good Luck!

Explore, analyze, and visualize your data thoroughly to derive meaningful insights!